



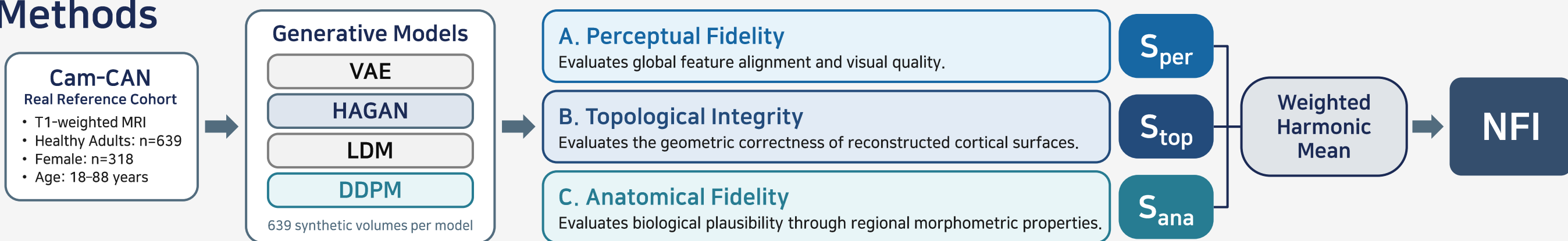
Beyond Visual Realism: The Neuro-Fidelity Index for Validating Geometric Integrity in Synthetic Brain MRI

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Introduction

- Diffusion models now set the state of the art for high-resolution brain MRI synthesis and are widely adopted to relieve data scarcity in neuroimaging [1, 2].
- FID [3] compares global feature distributions; PSNR, SSIM and Dice do not explicitly validate reconstructed surface geometry. Surface defects can compromise morphometry and parcellation [4].
- This study proposes the **Neuro-Fidelity Index (NFI)**, unifying perceptual sharpness, topological continuity and parcellation consistency into one interpretable score.
- A weighted harmonic mean makes it an **architectural bottleneck detector**: collapse in any one dimension is heavily penalized, exposing failures FID leaves invisible.

Methods



The Unified Neuro-Fidelity Index

$$NFI = 3 / (S_{per}^{-1} + S_{top}^{-1} + S_{ana}^{-1})$$

The weighted harmonic mean acts as an architectural bottleneck detector: catastrophic failure in any structural dimension is heavily penalized in the final score.

A. Perceptual Fidelity

$$S_{per} = \exp(-FID / \gamma_{per}), \gamma_{per} = 0.02$$

FID compares real and synthetic feature vectors from a pre-trained Inception-v3 network [3]. A normalized exponential decay maps FID to [0, 1], calibrated to the empirical within-cohort baseline variance.

B. Topological Integrity

$$S_{top} = \exp(-W(\chi_{syn}, \chi_{real}) / \gamma_{top})$$

Euler characteristic $\chi = V - E + F$ is compared with the real reference distribution using 1-Wasserstein distance. The normative baseline is 32.4 ± 14.5 ; $\gamma_{top} = \sigma(\chi_{real}) = 14.5$. A genus-0 closed hemispheric mesh has $\chi = 2$.

C. Anatomical Fidelity

$$S_{ana} = (s_{vol} + s_{area} + s_{thick}) / 3$$

$$s_k = \exp(-W_k / \gamma_k)$$

152 morphological features: 68 cortical thickness and 68 cortical surface area measures from Desikan-Killiany parcellation [9], plus 16 subcortical volumes from FreeSurfer's probabilistic atlas [10]. Category scores are averaged.

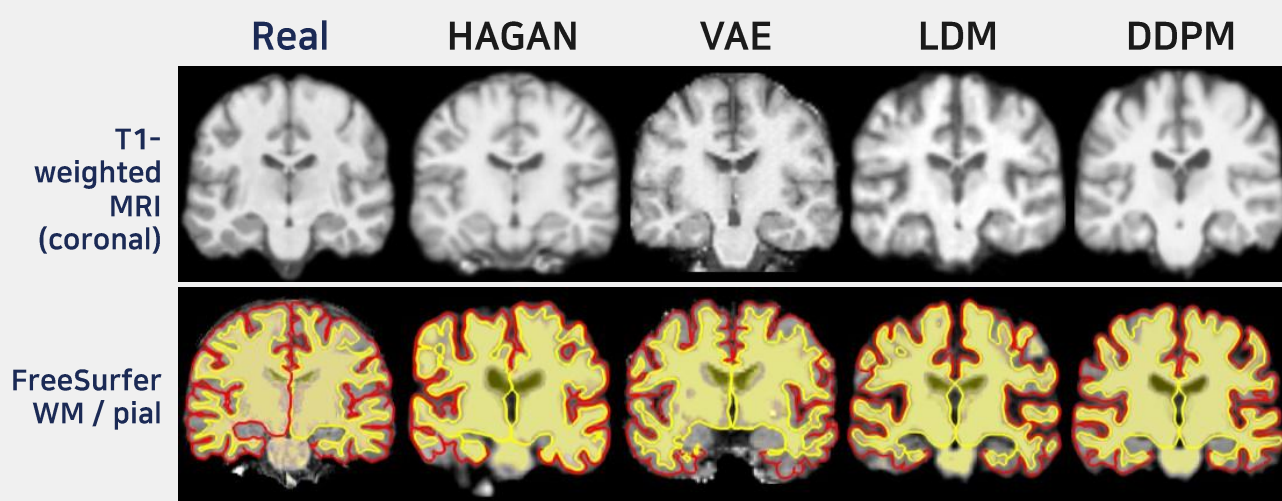
Experiments

- Reference cohort.** 639 cognitively healthy Cam-CAN participants (318 females; age 18-88 years) establish a baseline for normative anatomical variance [5].
- Preprocessing.** Skull removal, bias field correction, affine registration to MNI152 at 2 mm isotropic resolution, and FreeSurfer intensity normalization; volumes cropped from 91×109×91 to 72×96×72 voxels [6-8].
- Generative models.** 3D-VAE (four 3D convolutional layers; 256-dimensional latent bottleneck) [11], HAGAN (hierarchical discriminator) [12], LDM (pre-trained 3D-VAE; compression factor 4) [2], and image-space DDPM [1]. Diffusion models use a 3D U-Net backbone with residual blocks and multi-head spatial attention [13].
- Evaluation.** Each model synthesized 639 volumes, exported in NIfTI format and processed using the same automated FreeSurfer recon-all pipeline as the ground-truth data, without manual intervention [8]. Models were implemented in PyTorch and MONAI and trained on an NVIDIA RTX 6000 GPU.

Results

Model	FID ↓	S_{per} ↑	S_{top} ↑	S_{ana} ↑	Arithmetic	Geometric	NFI ↑
VAE	0.0273	0.255	0.014	0.697	0.322	0.089	0.040
HAGAN	0.0158	0.454	0.040	0.837	0.444	0.197	0.106
LDM	0.0024	0.887	0.001	0.867	0.585	0.027	0.002
DDPM	0.0020	0.905	0.735	0.880	0.840	0.835	0.833

- Perceptual paradox:** LDM achieves high perceptual (0.887) and anatomical (0.867) similarity, but severe topological collapse (0.001) yields a reported NFI of 0.002.
- Geometric discontinuities:** white/pial surface boundary errors and mesh fragmentation reveal structural failures in LDM-generated outputs.
- Balanced performance:** pixel-space DDPM achieves the highest NFI (0.833), with perceptual, topological, and anatomical scores of 0.905, 0.735, and 0.880.
- Visual realism does not guarantee geometric integrity. Reported NFI: LDM 0.002 vs. pixel-space DDPM 0.833.**



Figs. 1-2. Real and synthetic T1-weighted MRI, with white matter (yellow) and pial surface (red) boundaries.

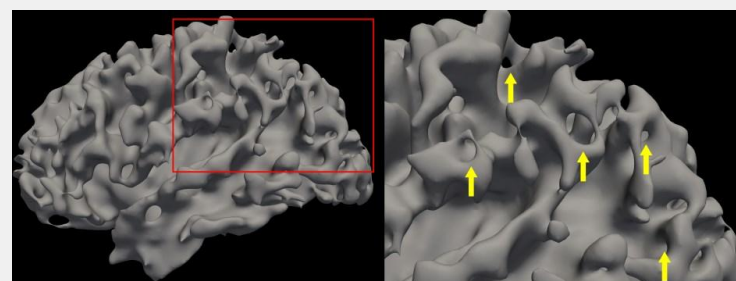


Fig. 3. Topological defects in LDM-generated cortical surfaces revealed by FreeSurfer surface reconstructions.

Conclusions

- NFI integrates perceptual, topological, and anatomical validation into a unified, interpretable benchmarking metric for synthetic brain MRI.
- The weighted harmonic mean acts as an architectural bottleneck detector, preventing structural failures from being masked by high perceptual similarity.
- The perceptual paradox separates LDM (NFI 0.002) from pixel-space DDPM (0.833). Model rankings remain stable in the reported weight sensitivity analysis.

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